

E-commerce Sales Data Cleaning

Service: Excel Data Cleaning & Quality Reporting

Tool: Microsoft Excel

Project type: Portfolio demonstration using sample data

Project overview

This project demonstrates how raw e-commerce sales data can be cleaned, standardized, and organized into a structured dataset suitable for business reporting.

Challenges identified

- Duplicate order record
- Missing quantity value
- Inconsistent text formatting
- Need for clear data-quality reporting

Work performed

- Identified and removed a duplicate record from the cleaned dataset.
- Standardized customer and region text values.
- Preserved missing values rather than guessing or fabricating data.
- Added data-quality status flags.
- Created a cleaning log documenting issues and actions.
- Built an automated summary of data-quality checks.

Figure 1. Cleaned dataset and data-quality summary

The screenshot displays an Excel spreadsheet titled 'Ecommerce_Sales_Data_Cleaning_Portfolio.xlsx'. The main data table, 'SALES DATA - CLEANED DATASET', contains 10 rows of sales data. The columns are: ORDER ID, Customer, Product, Category, Date, Quantity, Price, Revenue, Cost, Profit, Region, and Data Quality Status. The data includes various products like Laptops, Mice, Keyboards, Chairs, and Monitors, categorized under Electronics and Furniture. The 'Data Quality Status' column shows 'OK' for most records, with one 'CHECK' status for a record with a missing quantity. A 'DATA QUALITY SUMMARY' box in the bottom right corner provides a summary of the cleaning process: TOTAL ORDERS (9), DUPLICATE ORDERS REMOVED (1), MISSING QUANTITY (1), and RECORDS REQUIRING REVIEW (1). The spreadsheet also shows a 'CLEANING_LOG' tab and a 'RAW_DATA' tab at the bottom.

ORDER ID	Customer	Product	Category	Date	Quantity	Price	Revenue	Cost	Profit	Region	Data Quality Status
1001	Rahul Sharma	Laptop	Electronics	01-09-2026	2	50000	100000	80,000.00	20,000.00	South	OK
1002	Priya	Mouse	Electronics	02-09-2026	3	800	2400	1,500.00	900.00	South	OK
1003	Amit	Keyboard	Electronics	03-09-2026	1	1500	1500	1,000.00	500.00	North	OK
1004	Rahul Sharma	Chair	Furniture	04-09-2026	5000	5000	3,500.00			West	CHECK
1005	Priya	Mouse	Electronics	05-09-2026	2	800	1600	1,000.00	600.00	South	OK
1006	Neha	Desk	Furniture	06-09-2026	1	7000	7000	5,000.00	2,000.00	East	OK
1007	Arun	Monitor	Electronics	07-09-2026	2	12000	24000	18,000.00	6,000.00	West	OK
1008	Kavya	Chair	Furniture	08-09-2026	1	5000	5000	3,500.00	1,500.00	South	OK
1009	Rahul Sharma	Laptop	Electronics	09-09-2026	1	50000	50000	40,000.00	10,000.00	South	OK

DATA QUALITY SUMMARY

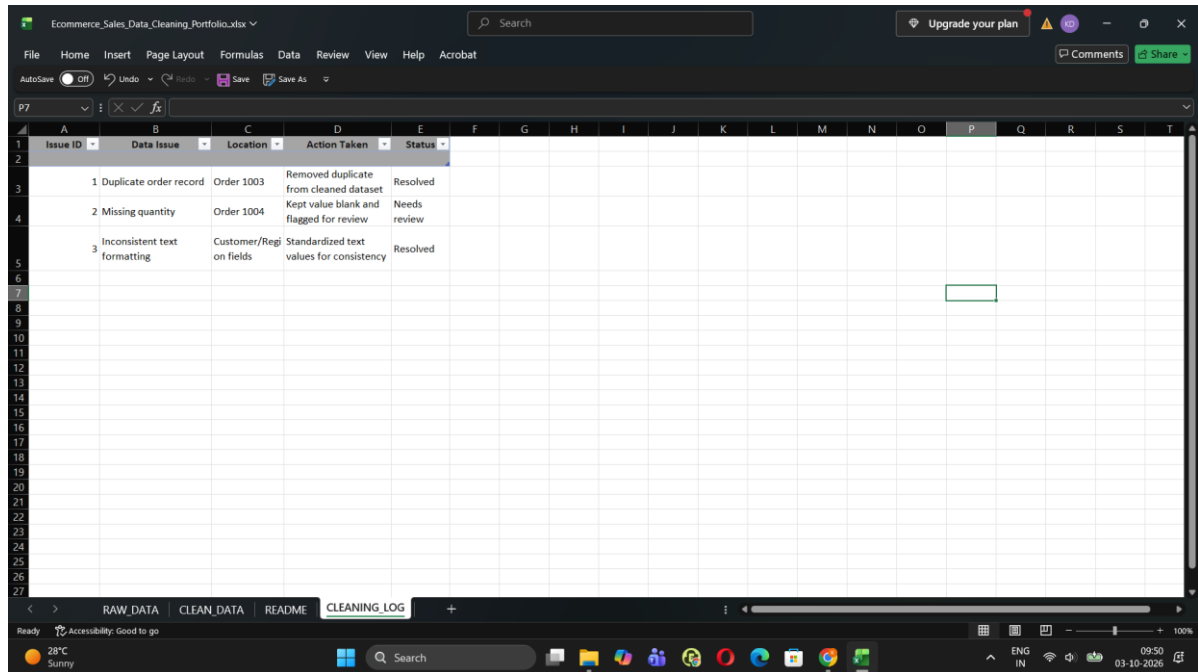
Metric	Count
TOTAL ORDERS	9
DUPLICATE ORDERS REMOVED	1
MISSING QUANTITY	1
RECORDS REQUIRING REVIEW	1

Deliverables

- Original raw dataset

- Cleaned dataset
- Cleaning log
- Data-quality summary
- README sheet

Figure 2. Documented data-cleaning log



Issue ID	Data Issue	Location	Action Taken	Status
1	Duplicate order record	Order 1003	Removed duplicate from cleaned dataset	Resolved
2	Missing quantity	Order 1004	Kept value blank and flagged for review	Needs review
3	Inconsistent text formatting	Customer/Registration fields	Standardized text values for consistency	Resolved

Outcome

The workbook contains 9 cleaned records, with 1 duplicate removed and 1 record flagged for review because its quantity is missing.

Note: This is a sample portfolio project, not work completed for an actual client. The dataset is used for demonstration purposes.

Ecommerce_Sales_Data_Cleaning_Portfolio.xlsx

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RAW_DATACLEAN_DATAREADMECLEANING_LOG

ReadyAccessibility: Good to go

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ORDER ID	Customer	Product	Category	Date	Quantity	Price	Revenue	Cost	Profit	Region	Data Quality Status
1001	Rahul Sharma	Laptop	Electronics	01-09-2026	2	50000.00	100000.00	80000.00	20000.00	South	OK
1002	Priya	Mouse	Electronics	02-09-2026	3	800.00	2400.00	1500.00	900.00	South	OK
1003	Amit	Keyboard	Electronics	03-09-2026	1	1500.00	1500.00	1000.00	500.00	North	OK
1004	Rahul Sharma	Chair	Furniture	04-09-2026	5000.00		3500.00			West	REVIEW - Missing Quantity
1005	Priya	Mouse	Electronics	05-09-2026	2	800.00	1600.00	1000.00	600.00	South	OK
1006	Neha	Desk	Furniture	06-09-2026	1	7000.00	7000.00	5000.00	2000.00	East	OK
1007	Arun	Monitor	Electronics	07-09-2026	2	12000.00	24000.00	18000.00	6000.00	West	OK
1008	Kavya	Chair	Furniture	08-09-2026	1	5000.00	5000.00	3500.00	1500.00	South	OK
1009	Rahul Sharma	Laptop	Electronics	09-09-2026	1	50000.00	50000.00	40000.00	10000.00	South	OK

DATA QUALITY SUMMARY
TOTAL ORDERS9
DUPLICATE ORDERS REMOVED1
MISSING QUANTITY1
RECORDS REQUIRING REVIEW1

RAW_DATACLEAN_DATAREADMECLEANING_LOG

ReadyAccessibility: Good to go